JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Differential Equations

JEE Main Mathematics · Previous Year Questions with Answer Key

250	181	2002 – 2026	21	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	179 questions
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1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 179 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0, x \in (0,1), \lim_{x\to 1^-} y(x) = 1$. Then $y\left(\frac{1}{2}\right)$ equals :

(A) $3 - \frac{\pi}{\sqrt{3}}$ (B) $4 - \sqrt{3}\pi$ (C) $4 - \frac{2\pi}{\sqrt{3}}$ (D) $3 - \frac{\pi}{2\sqrt{3}}$

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in \mathbf{R}$ and $f(0) \neq 0$. Let $g : [1,\infty) \to \mathbf{R}$ be a differentiable function such that

$x^2 g(x) = \int_1^x \left(\mathrm{t}^2 f(\mathrm{t}) - \mathrm{tg}(\mathrm{t})\right)dt$

Then $g(2)$ is equal to :

(A) $\frac{13}{8}$ (B) $\frac{11}{16}$ (C) $\frac{15}{32}$ (D) $\frac{17}{64}$

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Let $f : [1,\infty) \to \mathbf{R}$ be a differentiable function defined as $f(x) = \int_1^x f(\mathrm{t})\mathrm{dt} + (1-x)\left(\log_{\mathrm{e}} x - 1\right) + \mathrm{e}$.

Then the value of $f(f(1))$ is :

(A) $(1 + \mathrm{e}^{\mathrm{e}})$ (B) $(1 + \mathrm{e})$

(C) $(1 + \mathrm{e} + \mathrm{e}^{\mathrm{e}})$ (D) $1 + 2e$

Q4. JEE Main 2026 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation :

$\frac{dy}{dx} + \left(\frac{6x^2 + \left(3x^2 + 2x^3 + 4\right)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$

$x \in (-1,2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha\left(2 + e^{-2}\right)$, then α is equal to :

(A) $\frac{13}{8}$ (B) $\frac{6}{13}$ (C) $\frac{12}{13}$ (D) $\frac{13}{12}$

Q5. JEE Main 2026 (Online) 4th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \left(1 + x + x^2\right)\left(1 - y + y^2\right), y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to

(A) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$ (B) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

(C) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$ (D) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Let $x = x(y)$ be the solution of the differential equation $2y^2\frac{dx}{dy} - 2xy + x^2 = 0, y > 1, x(e) = e$.

Then $x(e^2)$ is equal to:

(A) $\frac{3}{2}e^2$ (B) $\frac{2}{3}e^2$ (C) e^2 (D) $2e^2$

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

If the curve $y = f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $dy = y(2 + \log_e x)dx, x > 0$, then $f(e)$ is equal to :

(A) e^e (B) e^{e^2} (C) e^{2e} (D) e^{2^e}

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation

$(1 + \sin x)\frac{dy}{dx} + (y + 1)\cos x = 0,\ y(0) = 0$. If the curve $y = y(x)$ passes through the point $\left(\alpha, \frac{-1}{2}\right)$,

then a value of α is:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2\cot x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then

$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

(A) -3π (B) 3π (C) $-\pi$ (D) π

Q10. JEE Main 2026 (Online) 28th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, x \neq 0$.

If $y(2) = 0$, then $\tan(y(1))$ is equal to

(A) $-\frac{7}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{7}{4}$

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$x^4\,dy + \left(4x^3y + 2\sin x\right)dx = 0, x > 0, y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

(A) 92 (B) 72 (C) 64 (D) 81

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

If $y = y(x)$ satisfies the differential equation $16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})\cos y\,dy = (1+2\sin y)dx, x > 0$ and $y(256) = \frac{\pi}{2}, y(49) = \alpha$, then $2\sin\alpha$ is equal to :

(A) $2\sqrt{2}-1$ (B) $\sqrt{2}-1$ (C) $2(\sqrt{2}-1)$ (D) $3(\sqrt{2}-1)$

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2+y^2}dx, x > 0, y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to

(A) 6 (B) 4 (C) 1 (D) 2

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\sec x\frac{dy}{dx} - 2y = 2 + 3\sin x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

(A) $-\frac{5}{2}$ (B) $-3\sqrt{2}-7$ (C) $-\frac{5}{4}$ (D) $-3\sqrt{3}-7$

Q15. JEE Main 2026 (Online) 21st January Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation $(1+x^2)dy + (y - \tan^{-1}x)dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

(A) $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$ (B) $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$

(C) $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$ (D) $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

Q16. JEE Main 2025 (Online) 8th April Evening Shift

Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in \mathbb{R}$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}}g(f(f(x))) - \frac{y}{\sqrt{x}}\right), y(0) = 0$, then $y(1)$ is

(A) $\frac{1-e^3}{e^4}$ (B) $\frac{e-1}{e^4}$ (C) $\frac{1-e^2}{e^4}$ (D) $\frac{2e-1}{e^3}$

Q17. JEE Main 2025 (Online) 7th April Evening Shift

Let y = y(x) be the solution of the differential equation $(x^2+1)y' - 2xy = (x^4+2x^2+1)\cos x$, $y(0) = 1$. Then $\int_{-3}^{3} y(x)\,dx$ is :

(A) 36 (B) 24 (C) 18 (D) 30

Q18. JEE Main 2025 (Online) 7th April Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation

$x\left(x^2+e^x\right)dy+\left(\mathrm{e}^x(x-2)y-x^3\right)\mathrm{d}x=0, x>0$, passing through the point $(1,0)$. Then $y(2)$ is equal to :

(A) $\frac{2}{2+e^2}$ (B) $\frac{4}{4-e^2}$ (C) $\frac{4}{4+e^2}$ (D) $\frac{2}{2-e^2}$

Q19. JEE Main 2025 (Online) 4th April Evening Shift

If a curve $y = y(x)$ passes through the point $\left(1,\frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4\cot y-\mathrm{e}^x\operatorname{cosec} y\right)\frac{\mathrm{d}x}{\mathrm{d}y}=x^5, x\geq 1$, then at $x = 2$, the value of $\cos y$ is :

(A) $\frac{2\mathrm{e}^2+\mathrm{e}}{64}$ (B) $\frac{2\mathrm{e}^2-\mathrm{e}}{64}$

(C) $\frac{2\mathrm{e}^2-\mathrm{e}}{128}$ (D) $\frac{2\mathrm{e}^2+\mathrm{e}}{128}$

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx}+3\left(\tan^2 x\right)y+3y=\sec^2 x, y(0)=\frac{1}{3}+e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{4}{3}$ (B) $\frac{2}{3}+e^3$ (C) $\frac{4}{3}+e^3$ (D) $\frac{2}{3}$

Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Let g be a differentiable function such that $\int_0^x g(t)dt = x-\int_0^x tg(t)dt, x\geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx}-y\tan x=2(x+1)\sec x g(x), x\in\left[0,\frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to

(A) $\frac{4\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{2\pi}{3\sqrt{3}}$ (D) $\frac{4\pi}{3\sqrt{3}}$

Q22. JEE Main 2025 (Online) 29th January Evening Shift

If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx}+(\tan x)y=\frac{2+\sec x}{(1+2\sec x)^2}$, $x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right), f\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

(A) $\frac{5-\sqrt{3}}{2\sqrt{2}}$ (B) $\frac{4-\sqrt{2}}{14}$

(C) $\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$ (D) $\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

Q23. JEE Main 2025 (Online) 29th January Morning Shift

Let *y = y(x)* be the solution of the differential equation :

$\cos x(\log_e(\cos x))^2dy+(\sin x-3y\sin x\log_e(\cos x))dx=0$, x ∈ (0, $\frac{\pi}{2}$). If $y(\frac{\pi}{4}) = -\frac{1}{\log_e 2}$, then $y(\frac{\pi}{6})$ is equal to :

(A) $\frac{2}{\log_e(3)-\log_e(4)}$ (B) $-\frac{1}{\log_e(4)}$

(C) $\frac{1}{\log_e(4)-\log_e(3)}$ (D) $\frac{1}{\log_e(3)-\log_e(4)}$

Q24. JEE Main 2025 (Online) 28th January Morning Shift

Let for some function $y = f(x), \int_0^x tf(t)dt = x^2 f(x), x > 0$ and $f(2) = 3$. Then $f(6)$ is equal to

(A) 1 (B) 6 (C) 2 (D) 3

Q25. JEE Main 2025 (Online) 24th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + \left(1+x^2\right)dy = 0, y(0) = 0$. Then $y(\sqrt{3})$ is equal to

(A) $\frac{5\sqrt{3}}{2}$ (B) $\sqrt{\frac{15}{2}}$ (C) $\sqrt{\frac{14}{3}}$ (D) $2\sqrt{2}$

Q26. JEE Main 2025 (Online) 23rd January Evening Shift

Let $x = x(y)$ be the solution of the differential equation $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right), y > 0$ and $x(1) = \frac{\pi}{2}$. Then $\cos(x(2))$ is equal to :

(A) $2\left(\log_e 2\right) - 1$ (B) $1 - 2(\log_e 2)^2$

(C) $1 - 2\left(\log_e 2\right)$ (D) $2(\log_e 2)^2 - 1$

Q27. JEE Main 2025 (Online) 23rd January Morning Shift

Let a curve $y = f(x)$ pass through the points $(0, 5)$ and $(\log_e 2, k)$. If the curve satisfies the differential equation $2(3+y)e^{2x}dx - \left(7+e^{2x}\right)dy = 0$, then k is equal to

(A) 32 (B) 8 (C) 4 (D) 16

Q28. JEE Main 2025 (Online) 22nd January Evening Shift

If $x = f(y)$ is the solution of the differential equation $\left(1+y^2\right) + \left(x - 2e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $f(0) = 1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

(A) $e^{\pi/4}$ (B) $e^{\pi/12}$ (C) $e^{\pi/6}$ (D) $e^{\pi/3}$

Q29. JEE Main 2025 (Online) 22nd January Morning Shift

Let $x = x(y)$ be the solution of the differential equation $y^2\,dx + \left(x - \frac{1}{y}\right)dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is :

(A) $\frac{3}{2} + e$ (B) $\frac{1}{2} + e$ (C) $3 + e$ (D) $3 - e$

Q30. JEE Main 2025 (Online) 22nd January Morning Shift

Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbf{R}$. Then $\sum_{n=1}^{100} \log_e f(n)$ is equal to :

(A) 2406 (B) 5220 (C) 2525 (D) 2384

Q31. JEE Main 2025 (Online) 22nd January Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbf{R}$. If $f'(0) = 4\mathrm{a}$ and f satisfies $f''(x) - 3\mathrm{a}f'(x) - f(x) = 0, \mathrm{a} > 0$, then the area of the region $\mathrm{R} = \{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ is :

(A) $\mathrm{e}^2 - 1$ (B) $e^4 + 1$ (C) $\mathrm{e}^2 + 1$ (D) $e^4 - 1$

Q32. JEE Main 2024 (Online) 9th April Evening Shift

Let $\int_0^x \sqrt{1 - (y'(t))^2}\,dt = \int_0^x y(t)dt, 0 \le x \le 3, y \ge 0, y(0) = 0$. Then at $x = 2, y'' + y + 1$ is equal to

(A) $\sqrt{2}$ (B) 2 (C) 1/2 (D) 1

Q33. JEE Main 2024 (Online) 9th April Morning Shift

The solution of the differential equation $(x^2 + y^2)\mathrm{d}x - 5xy\,\mathrm{d}y = 0, y(1) = 0$, is :

(A) $\left|x^2 - 4y^2\right|^5 = x^2$ (B) $\left|x^2 - 2y^2\right|^6 = x$ (C) $\left|x^2 - 2y^2\right|^5 = x^2$ (D) $\left|x^2 - 4y^2\right|^6 = x$

Q34. JEE Main 2024 (Online) 9th April Morning Shift

The solution curve, of the differential equation $2y\frac{\mathrm{d}y}{\mathrm{d}x} + 3 = 5\frac{\mathrm{d}y}{\mathrm{d}x}$, passing through the point $(0, 1)$ is a conic, whose vertex lies on the line :

(A) $2x + 3y = -9$ (B) $2x + 3y = -6$ (C) $2x + 3y = 9$ (D) $2x + 3y = 6$

Q35. JEE Main 2024 (Online) 8th April Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\sec y\frac{\mathrm{d}y}{\mathrm{d}x} + 2x\sin y = x^3\cos y, y(1) = 0$. Then $y(\sqrt{3})$ is equal to:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q36. JEE Main 2024 (Online) 8th April Morning Shift

Let $f(x)$ be a positive function such that the area bounded by $y = f(x), y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is

(A) $(8e^x + 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ (B) $(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$
(C) $(8e^x - 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ (D) $(8e^x - 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Q37. JEE Main 2024 (Online) 8th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x}dx + \cos^2 x(1 + e^{2\tan x})dy = 0, y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to

(A) $\frac{1}{e^2}$ (B) $\frac{2}{e^2}$ (C) $\frac{2}{e}$ (D) $\frac{1}{e}$

Q38. JEE Main 2024 (Online) 6th April Evening Shift

Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x-\beta y+2}{\beta x-2\alpha y-(\beta\gamma-4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is :

(A) $\sqrt{17}$ (B) 2 (C) $\frac{\sqrt{17}}{2}$ (D) $\frac{1}{2}$

Q39. JEE Main 2024 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ and $y\left(e^{-1}\right) = 0$. Then, $y(e)$ is equal to

(A) $-\frac{3}{e}$ (B) $-\frac{3}{2e}$ (C) $-\frac{2}{3e}$ (D) $-\frac{2}{e}$

Q40. JEE Main 2024 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx} + y = e^{\tan^{-1}x}$, $y(1) = 0$. Then $y(0)$ is

(A) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ (B) $\frac{1}{2}\left(1-e^{\pi/2}\right)$
(C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ (D) $\frac{1}{2}\left(e^{\pi/2}-1\right)$

Q41. JEE Main 2024 (Online) 5th April Evening Shift

The differential equation of the family of circles passing through the origin and having centre at the line $y = x$ is :

(A) $\left(x^2-y^2+2xy\right)dx = \left(x^2-y^2+2xy\right)dy$ (B) $\left(x^2+y^2-2xy\right)dx = \left(x^2+y^2+2xy\right)dy$
(C) $\left(x^2+y^2+2xy\right)dx = \left(x^2+y^2-2xy\right)dy$ (D) $\left(x^2-y^2+2xy\right)dx = \left(x^2-y^2-2xy\right)dy$

Q42. JEE Main 2024 (Online) 5th April Morning Shift

If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + 2y = \sin(2x), y(0) = \frac{3}{4}$, then $y\left(\frac{\pi}{8}\right)$ is equal to :

(A) $e^{-\pi/8}$ (B) $e^{\pi/4}$ (C) $e^{-\pi/4}$ (D) $e^{\pi/8}$

Q43. JEE Main 2024 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x^2+4)^2dy + (2x^3y+8xy-2)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to

(A) 2π (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{16}$ (D) $\frac{\pi}{32}$

Q44. JEE Main 2024 (Online) 4th April Morning Shift

If the solution $y = y(x)$ of the differential equation
$(x^4+2x^3+3x^2+2x+2)dy - (2x^2+2x+3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to :

(A) $-\frac{\pi}{12}$ (B) $\frac{\pi}{2}$ (C) 0 (D) $\frac{\pi}{4}$

Q45. JEE Main 2024 (Online) 1st February Evening Shift

Let α be a non-zero real number. Suppose $f : \mathbf{R} \to \mathbf{R}$ is a differentiable function such that $f(0) = 2$ and $\lim\limits_{x \to -\infty} f(x) = 1$. If $f'(x) = \alpha f(x) + 3$, for all $x \in \mathbf{R}$, then $f(-\log_e 2)$ is equal to :

(A) 7 (B) 9 (C) 3 (D) 5

Q46. JEE Main 2024 (Online) 1st February Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1, y(0) = 1.$

Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals :

(A) $\frac{4}{4+\sqrt{e}}$ (B) $\frac{3}{3-\sqrt{e}}$ (C) $\frac{2}{1+\sqrt{e}}$ (D) $\frac{1}{2-\sqrt{e}}$

Q47. JEE Main 2024 (Online) 31st January Evening Shift

The temperature $T(t)$ of a body at time $t = 0$ is 160°F and it decreases continuously as per the differential equation $\frac{dT}{dt} = -K(T - 80)$, where K is a positive constant. If $T(15) = 120^\circ\text{F}$, then $T(45)$ is equal to

(A) 90° F (B) 85° F (C) 80° F (D) 95° F

Q48. JEE Main 2024 (Online) 31st January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}, x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then, $y\left(\frac{\pi}{3}\right)$ is

(A) $\sqrt{3}\,(2 + \log_e 3)$ (B) $\sqrt{3}\,(1 + 2\log_e 3)$

(C) $\sqrt{3}\left(2 + \log_e \sqrt{3}\right)$ (D) $\frac{\sqrt{3}}{2}(2 + \log_e 3)$

Q49. JEE Main 2024 (Online) 31st January Morning Shift

The solution curve of the differential equation $y\frac{dx}{dy} = x\,(\log_e x - \log_e y + 1), x > 0, y > 0$ passing through the point $(e, 1)$ is

(A) $\left|\log_e \frac{y}{x}\right| = y^2$ (B) $\left|\log_e \frac{y}{x}\right| = x$

(C) $\left|\log_e \frac{x}{y}\right| = y$ (D) $2\left|\log_e \frac{x}{y}\right| = y + 1$

Q50. JEE Main 2024 (Online) 30th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\sec x\, dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to:

(A) $2\{\sin(2) + 1\}$ (B) 2 (C) 1 (D) $2\{1 - \sin(2)\}$

Q51. JEE Main 2024 (Online) 29th January Evening Shift

If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to

(A) 12 (B) 9 (C) 4 (D) 3

Q52. JEE Main 2024 (Online) 29th January Morning Shift

A function $y = f(x)$ satisfies $f(x)\sin 2x + \sin x - \left(1+\cos^2 x\right)f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to

(A) 2 (B) 1 (C) -1 (D) 0

Q53. JEE Main 2024 (Online) 27th January Evening Shift

If $y = y(x)$ is the solution curve of the differential equation $\left(x^2-4\right)\mathrm{d}y - \left(y^2-3y\right)\mathrm{d}x = 0, x > 2, y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

(A) $\frac{3}{1+(8)^{1/4}}$ (B) $\frac{3}{1-(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1+2\sqrt{2}}$

Q54. JEE Main 2024 (Online) 27th January Morning Shift

Let $x = x(\mathrm{t})$ and $y = y(\mathrm{t})$ be solutions of the differential equations $\frac{\mathrm{d}x}{\mathrm{dt}} + \mathrm{a}x = 0$ and $\frac{\mathrm{d}y}{\mathrm{dt}} + \mathrm{b}y = 0$ respectively, $\mathrm{a}, \mathrm{b} \in \mathbf{R}$. Given that $x(0) = 2; y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(\mathrm{t}) = y(\mathrm{t})$, is :

(A) $\log_{\frac{2}{3}} 2$ (B) $\log_{\frac{4}{3}} 2$ (C) $\log_4 3$ (D) $\log_3 4$

Q55. JEE Main 2023 (Online) 15th April Morning Shift

Let $x = x(y)$ be the solution of the differential equation

$$2(y+2)\log_e(y+2)dx + (x+4-2\log_e(y+2))dy = 0, y > -1$$

with $x\left(e^4-2\right) = 1$. Then $x\left(e^9-2\right)$ is equal to :

(A) $\frac{4}{9}$ (B) $\frac{32}{9}$ (C) $\frac{10}{3}$ (D) 3

Q56. JEE Main 2023 (Online) 13th April Morning Shift

Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at

(A) no point (B) two points

(C) infinite number of points (D) one point

Q57. JEE Main 2023 (Online) 12th April Morning Shift

Let $y = y(x), y > 0$, be a solution curve of the differential equation $\left(1+x^2\right)\mathrm{d}y = y(x-y)\mathrm{d}x$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then

(A) $e^{\beta^{-1}} = e^{-2}(3+2\sqrt{2})$

(B) $e^{3\beta^{-1}} = e(5+\sqrt{2})$

(C) $e^{3\beta^{-1}} = e(3+2\sqrt{2})$

(D) $e^{\beta^{-1}} = e^{-2}(5+\sqrt{2})$

Q58. JEE Main 2023 (Online) 11th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x(x^5+1)}y = \frac{(x^5+1)^2}{x^7}, x > 0$. If $y(1) = 2$, then $y(2)$ is equal to :

(A) $\frac{693}{128}$

(B) $\frac{697}{128}$

(C) $\frac{637}{128}$

(D) $\frac{679}{128}$

Q59. JEE Main 2023 (Online) 11th April Morning Shift

Let $y = y(x)$ be a solution curve of the differential equation

$\left(1-x^2y^2\right)dx = ydx + xdy$.

If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α is :

(A) $\frac{1+3e^2}{2(3e^2-1)}$

(B) $\frac{3e^2}{2(3e^2-1)}$

(C) $\frac{1-3e^2}{2(3e^2+1)}$

(D) $\frac{3e^2}{2(3e^2+1)}$

Q60. JEE Main 2023 (Online) 10th April Morning Shift

Let f be a differentiable function such that $x^2f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$. Then $18f(3)$ is equal to :

(A) 160

(B) 210

(C) 150

(D) 180

Q61. JEE Main 2023 (Online) 6th April Evening Shift

If the solution curve $f(x, y) = 0$ of the differential equation

$(1+\log_e x)\frac{dx}{dy} - x\log_e x = e^y, x > 0$,

passes through the points $(1, 0)$ and $(\alpha, 2)$, then α^α is equal to :

(A) $e^{\sqrt{2}e^2}$

(B) $e^{2e^{\sqrt{2}}}$

(C) e^{e^2}

(D) e^{2e^2}

Q62. JEE Main 2023 (Online) 1st February Evening Shift

Let $\alpha x = \exp\left(x^\beta y^\gamma\right)$ be the solution of the differential equation $2x^2y\,\mathrm{d}y - \left(1-xy^2\right)\mathrm{d}x = 0, x > 0, y(2) = \sqrt{\log_e 2}$. Then $\alpha+\beta-\gamma$ equals :

(A) 1

(B) 0

(C) 3

(D) -1

Q63. JEE Main 2023 (Online) 1st February Morning Shift

The area enclosed by the closed curve C given by the differential equation

$\frac{dy}{dx}+\frac{x+a}{y-2}=0, y(1)=0$ is 4π.

Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is :

(A) $\frac{4\sqrt{3}}{3}$ (B) $2\sqrt{3}$ (C) 2 (D) $\frac{2\sqrt{3}}{3}$

Q64. JEE Main 2023 (Online) 1st February Morning Shift

If $y=y(x)$ is the solution curve of the differential equation

$\frac{dy}{dx}+y\tan x=x\sec x, 0\le x\le\frac{\pi}{3}, y(0)=1$, then $y\left(\frac{\pi}{6}\right)$ is equal to

(A) $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$ (B) $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(C) $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$ (D) $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Q65. JEE Main 2023 (Online) 31st January Evening Shift

Let $y=y(x)$ be the solution of the differential equation

$\left(3y^2-5x^2\right)y\,dx+2x\left(x^2-y^2\right)dy=0$

such that $y(1)=1$. Then $\left|(y(2))^3-12y(2)\right|$ is equal to :

(A) 64 (B) $16\sqrt{2}$ (C) 32 (D) $32\sqrt{2}$

Q66. JEE Main 2023 (Online) 30th January Evening Shift

The solution of the differential equation

$\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right), y(1)=0$ is :

(A) $\log_e|x+y|+\frac{xy}{(x+y)^2}=0$ (B) $\log_e|x+y|-\frac{xy}{(x+y)^2}=0$

(C) $\log_e|x+y|+\frac{2xy}{(x+y)^2}=0$ (D) $\log_e|x+y|-\frac{2xy}{(x+y)^2}=0$

Q67. JEE Main 2023 (Online) 30th January Morning Shift

Let the solution curve $y=y(x)$ of the differential equation

$\frac{dy}{dx}-\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{(1+x^6)}}\right\}$ pass through the origin. Then $y(1)$ is equal to :

(A) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (B) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

(C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (D) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$

Q68. JEE Main 2023 (Online) 29th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\log_e x\frac{dy}{dx} + y = x^2\log_e x, (x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to

(A) $\frac{1+e^2}{2}$ (B) $\frac{1+e^2}{4}$ (C) $\frac{2+e^2}{2}$ (D) $\frac{4+e^2}{4}$

Q69. JEE Main 2023 (Online) 29th January Morning Shift

Let $y = f(x)$ be the solution of the differential equation $y(x+1)dx - x^2dy = 0, y(1) = e$. Then $\lim\limits_{x\to 0^+} f(x)$ is equal to

(A) e^2 (B) 0 (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Q70. JEE Main 2023 (Online) 25th January Evening Shift

Let $y = y(t)$ be a solution of the differential equation $\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$ where, $\alpha > 0, \beta > 0$ and $\gamma > 0$. Then $\lim\limits_{t\to\infty} y(t)$

(A) is 0 (B) is 1 (C) is -1 (D) does not exist

Q71. JEE Main 2023 (Online) 25th January Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right), x > 0, y(1) = 3$. Then $\frac{y^2(x)}{9}$ is equal to :

(A) $\frac{x^2}{5-2x^3(2+\log_e x^3)}$ (B) $\frac{x^2}{3x^3(1+\log_e x^2)-2}$

(C) $\frac{x^2}{7-3x^3(2+\log_e x^2)}$ (D) $\frac{x^2}{2x^3(2+\log_e x^3)-3}$

Q72. JEE Main 2023 (Online) 24th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)dx + 3xy\,dy = 0, y(1) = 1$. Then $6y^2(e)$ is equal to

(A) $\frac{3}{2}e^2$ (B) $3e^2$ (C) e^2 (D) $2e^2$

Q73. JEE Main 2023 (Online) 24th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $x^3dy + (xy - 1)dx = 0, x > 0, y\left(\frac{1}{2}\right) = 3 - e$. Then y (1) is equal to

(A) $2 - e$ (B) 3 (C) 1 (D) e

Q74. JEE Main 2022 (Online) 29th July Evening Shift

If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passes through the points $(2, 1)$ and $(k + 1, 2), k > 0$, then

(A) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2 + 1\right)$ (B) $\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2 + 1\right)$

(C) $2\tan^{-1}\left(\frac{1}{k+1}\right) = \log_e\left(k^2 + 2k + 2\right)$ (D) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(\frac{k^2+1}{k^2}\right)$

Q75. JEE Main 2022 (Online) 29th July Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y = \frac{(x+3)}{x+1}, x > -1$, which passes through the point $(0, 1)$. Then $y(1)$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Q76. JEE Main 2022 (Online) 29th July Morning Shift

Let the solution curve $y = y(x)$ of the differential equation $\left(1 + \mathrm{e}^{2x}\right)\left(\frac{\mathrm{d}y}{\mathrm{d}x} + y\right) = 1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim\limits_{x\to\infty} \mathrm{e}^{x} y(x)$ is equal to :

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

Q77. JEE Main 2022 (Online) 28th July Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{1/2}, x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}\, y(8)$ is equal to :

(A) $11 + 6\log_e 3$ (B) 19 (C) $12 - 2\log_e 3$ (D) $19 - 6\log_e 3$

Q78. JEE Main 2022 (Online) 28th July Morning Shift

Let the solution curve of the differential equation $x\,\mathrm{d}y = \left(\sqrt{x^2+y^2} + y\right)\mathrm{d}x, x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Q79. JEE Main 2022 (Online) 28th July Morning Shift

If $y = y(x), x \in (0, \pi/2)$ be the solution curve of the differential equation

$$\left(\sin^2 2x\right)\frac{dy}{dx} + \left(8\sin^2 2x + 2\sin 4x\right)y = 2\mathrm{e}^{-4x}(2\sin 2x + \cos 2x),$$

with $y(\pi/4) = \mathrm{e}^{-\pi}$, then $y(\pi/6)$ is equal to :

(A) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}\mathrm{e}^{2\pi/3}$

(C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Q80. JEE Main 2022 (Online) 27th July Morning Shift

Let $y = y_1(x)$ and $y = y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx} = x + y$, with $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then, the number of points of intersection of $y = y_1(x)$ and $y = y_2(x)$ is

(A) 0 (B) 1 (C) 2 (D) 3

Q81. JEE Main 2022 (Online) 26th July Evening Shift

Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$, $x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

(A) $\frac{\pi}{3} - \frac{1}{4}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Q82. JEE Main 2022 (Online) 26th July Morning Shift

If $\frac{dy}{dx} + 2y\tan x = \sin x$, $0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is :

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Q83. JEE Main 2022 (Online) 25th July Morning Shift

The slope of the tangent to a curve $C : y = y(x)$ at any point (x, y) on it is $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$. If C passes through the points $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :

(A) $\frac{3+\sqrt{2}}{3-\sqrt{2}}$ (B) $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$ (D) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Q84. JEE Main 2022 (Online) 25th July Morning Shift

The general solution of the differential equation $(x - y^2)dx + y(5x + y^2)dy = 0$ is :

(A) $(y^2 + x)^4 = C\left|(y^2 + 2x)^3\right|$ (B) $(y^2 + 2x)^4 = C\left|(y^2 + x)^3\right|$

(C) $\left|(y^2 + x)^3\right| = C(2y^2 + x)^4$ (D) $\left|(y^2 + 2x)^3\right| = C(2y^2 + x)^4$

Q85. JEE Main 2022 (Online) 29th June Evening Shift

If y = y(x) is the solution of the differential equation $(1 + e^{2x})\frac{dy}{dx} + 2(1 + y^2)e^x = 0$ and y (0) = 0, then $6\left(y'(0) + \left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ is equal to

(A) 2 (B) -2 (C) -4 (D) -1

Q86. JEE Main 2022 (Online) 29th June Morning Shift

Let the solution curve of the differential equation $x\frac{dy}{dx} - y = \sqrt{y^2 + 16x^2}$, $y(1) = 3$ be $y = y(x)$. Then y(2) is equal to:

(A) 15 (B) 11 (C) 13 (D) 17

Q87. JEE Main 2022 (Online) 28th June Evening Shift

Let x = x(y) be the solution of the differential equation

$2y\,e^{x/y^2}dx + \left(y^2 - 4xe^{x/y^2}\right)dy = 0$ such that x(1) = 0. Then, x(e) is equal to :

(A) $e\log_e(2)$ (B) $-e\log_e(2)$ (C) $e^2\log_e(2)$ (D) $-e^2\log_e(2)$

Q88. JEE Main 2022 (Online) 28th June Evening Shift

Let the slope of the tangent to a curve y = f(x) at (x, y) be given by $2\tan x(\cos x - y)$. If the curve passes through the point $\left(\frac{\pi}{4}, 0\right)$, then the value of $\int_0^{\pi/2} y\,dx$ is equal to :

(A) $(2-\sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (B) $2 - \frac{\pi}{\sqrt{2}}$

(C) $(2+\sqrt{2}) + \frac{\pi}{\sqrt{2}}$ (D) $2 + \frac{\pi}{\sqrt{2}}$

Q89. JEE Main 2022 (Online) 28th June Morning Shift

Let the solution curve $y = y(x)$ of the differential equation

$\left[\frac{x}{\sqrt{x^2-y^2}} + e^{\frac{y}{x}}\right]x\frac{dy}{dx} = x + \left[\frac{x}{\sqrt{x^2-y^2}} + e^{\frac{y}{x}}\right]y$

pass through the points (1, 0) and (2α, α), α > 0. Then α is equal to

(A) $\frac{1}{2}\exp\left(\frac{\pi}{6} + \sqrt{e} - 1\right)$ (B) $\frac{1}{2}\exp\left(\frac{\pi}{6} + e - 1\right)$

(C) $\exp\left(\frac{\pi}{6} + \sqrt{e} + 1\right)$ (D) $2\exp\left(\frac{\pi}{3} + \sqrt{e} - 1\right)$

Q90. JEE Main 2022 (Online) 28th June Morning Shift

Let y = y(x) be the solution of the differential equation $x(1-x^2)\frac{dy}{dx} + (3x^2y - y - 4x^3) = 0, x > 1$, with $y(2) = -2$. Then y(3) is equal to :

(A) -18 (B) -12 (C) -6 (D) -3

Q91. JEE Main 2022 (Online) 27th June Evening Shift

If the solution curve of the differential equation

$((\tan^{-1}y) - x)dy = (1+y^2)dx$ passes through the point (1, 0), then the abscissa of the point on the curve whose ordinate is tan(1), is

(A) 2e (B) $\frac{2}{e}$ (C) 2 (D) $\frac{1}{e}$

Q92. JEE Main 2022 (Online) 27th June Morning Shift

If $\frac{dy}{dx} + \frac{2^{x-y}(2^y-1)}{2^x-1} = 0$, x, y > 0, y(1) = 1, then y(2) is equal to :

(A) $2 + \log_2 3$ (B) $2 + \log_3 2$ (C) $2 - \log_3 2$ (D) $2 - \log_2 3$

Q93. JEE Main 2022 (Online) 26th June Evening Shift

If $y = y(x)$ is the solution of the differential equation

$x\frac{dy}{dx} + 2y = x\,e^x$, $y(1) = 0$ then the local maximum value

of the function $z(x) = x^2y(x) - e^x,\ x \in R$ is :

(A) 1 − e (B) 0 (C) $\frac{1}{2}$ (D) $\frac{4}{e} - e$

Q94. JEE Main 2022 (Online) 26th June Evening Shift

If the solution of the differential equation

$\frac{dy}{dx} + e^x\left(x^2 - 2\right)y = \left(x^2 - 2x\right)\left(x^2 - 2\right)e^{2x}$ satisfies $y(0) = 0$, then the value of y(2) is _____.

(A) -1 (B) 1 (C) 0 (D) e

Q95. JEE Main 2022 (Online) 25th June Evening Shift

If $y = y(x)$ is the solution of the differential equation

$2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then y(1) is equal to :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Q96. JEE Main 2022 (Online) 25th June Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(x+1)y' - y = e^{3x}(x+1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve $y = y(x)$ is :

(A) not a critical point (B) a point of local minima

(C) a point of local maxima (D) a point of inflection

Q97. JEE Main 2022 (Online) 25th June Morning Shift

If the solution curve $y = y(x)$ of the differential equation $y^2dx + (x^2 - xy + y^2)dy = 0$, which passes through the point (1, 1) and intersects the line $y = \sqrt{3}x$ at the point $(\alpha, \sqrt{3}\alpha)$, then value of $\log_e(\sqrt{3}\alpha)$ is equal to :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

Q98. JEE Main 2022 (Online) 24th June Morning Shift

If x = x(y) is the solution of the differential equation

$y\frac{dx}{dy} = 2x + y^3(y+1)e^y$, $x(1) = 0$; then x(e) is equal to :

(A) $e^3(e^e - 1)$ (B) $e^e(e^3 - 1)$ (C) $e^2(e^e + 1)$ (D) $e^e(e^2 - 1)$

Q99. JEE Main 2021 (Online) 1st September Evening Shift

If y = y(x) is the solution curve of the differential equation $x^2dy + \left(y - \frac{1}{x}\right)dx = 0$; x > 0 and y(1) = 1, then $y\left(\frac{1}{2}\right)$ is equal to :

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $3 + \frac{1}{\sqrt{e}}$

(C) 3 + e (D) $3 - e$

Q100. JEE Main 2021 (Online) 31st August Evening Shift

If $\frac{dy}{dx} = \frac{2^x y + 2^y.2^x}{2^x + 2^{x+y}\log_e 2}$, y(0) = 0, then for y = 1, the value of x lies in the interval :

(A) (1, 2) (B) $\left(\frac{1}{2}, 1\right]$ (C) (2, 3) (D) $\left(0, \frac{1}{2}\right]$

Q101. JEE Main 2021 (Online) 31st August Evening Shift

If $y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, x > 0, ϕ > 0, and y(1) = -1, then $\phi\left(\frac{y^2}{4}\right)$ is equal to :

(A) 4 ϕ (2) (B) 4ϕ (1) (C) 2 ϕ (1) (D) ϕ (1)

Q102. JEE Main 2021 (Online) 31st August Morning Shift

If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, y(0) = 1, then y(1) is equal to :

(A) $\log_2(2 + e)$ (B) $\log_2(1 + e)$ (C) $\log_2(2e)$ (D) $\log_2(1 + e^2)$

Q103. JEE Main 2021 (Online) 27th August Evening Shift

If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points (0, 1) and (2, β), then β is a root of the equation :

(A) $y^5 - 2y - 2 = 0$ (B) $2y^5 - 2y - 1 = 0$ (C) $2y^5 - y^2 - 2 = 0$ (D) $y^5 - y^2 - 1 = 0$

Q104. JEE Main 2021 (Online) 27th August Morning Shift

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = 2(y + 2\sin x - 5)x - 2\cos x$ such that y(0) = 7. Then y(π) is equal to :

(A) $2e^{\pi^2} + 5$ (B) $e^{\pi^2} + 5$ (C) $3e^{\pi^2} + 5$ (D) $7e^{\pi^2} + 5$

Q105. JEE Main 2021 (Online) 27th August Morning Shift

Let us consider a curve, y = f(x) passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point (x, f(x)) is given by f(x) + xf'(x) = x^2. Then :

(A) $x^2 + 2xf(x) - 12 = 0$ (B) $x^3 + xf(x) + 12 = 0$

(C) $x^3 - 3xf(x) - 4 = 0$ (D) $x^2 + 2xf(x) + 4 = 0$

Q106. JEE Main 2021 (Online) 26th August Evening Shift

Let y(x) be the solution of the differential equation

$2x^2\,dy + (e^y - 2x)dx = 0$, $x > 0$. If $y(e) = 1$, then $y(1)$ is equal to :

(A) 0 (B) 2 (C) $\log_e 2$ (D) $\log_e (2e)$

Q107. JEE Main 2021 (Online) 26th August Morning Shift

Let y = y(x) be a solution curve of the differential equation $(y+1)\tan^2 x\,dx + \tan x\,dy + y\,dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$. If $\lim\limits_{x\to 0+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is :

(A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4} - 1$ (C) $\frac{\pi}{4} + 1$ (D) $\frac{\pi}{4}$

Q108. JEE Main 2021 (Online) 27th July Evening Shift

Let y = y(x) be the solution of the differential

equation $(x - x^3)dy = (y + yx^2 - 3x^4)dx$, $x > 2$. If $y(3) = 3$, then $y(4)$ is equal to :

(A) 4 (B) 12 (C) 8 (D) 16

Q109. JEE Main 2021 (Online) 27th July Morning Shift

Let y = y(x) be solution of the differential equation

$\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with $y(0) = 0$.

If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha \log_e 2$, then the value of α is equal to :

(A) $-\frac{1}{4}$ (B) $\frac{1}{4}$ (C) 2 (D) $-\frac{1}{2}$

Q110. JEE Main 2021 (Online) 25th July Evening Shift

Let y = y(x) be the solution of the differential

equation $xdy = (y + x^3\cos x)dx$ with $y(\pi) = 0$, then $y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $\frac{\pi^2}{4} + \frac{\pi}{2}$ (B) $\frac{\pi^2}{2} + \frac{\pi}{4}$

(C) $\frac{\pi^2}{2} - \frac{\pi}{4}$ (D) $\frac{\pi^4}{4} - \frac{\pi}{2}$

Q111. JEE Main 2021 (Online) 25th July Morning Shift

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 1 + xe^{y-x}, -\sqrt{2} < x < \sqrt{2}, y(0) = 0$

then, the minimum value of $y(x), x \in \left(-\sqrt{2}, \sqrt{2}\right)$ is equal to :

(A) $\left(2-\sqrt{3}\right) - \log_e 2$ (B) $\left(2+\sqrt{3}\right) + \log_e 2$

(C) $\left(1+\sqrt{3}\right) - \log_e\left(\sqrt{3}-1\right)$ (D) $\left(1-\sqrt{3}\right) - \log_e\left(\sqrt{3}-1\right)$

Q112. JEE Main 2021 (Online) 22th July Evening Shift

Let y = y(x) be the solution of the differential equation $\text{cosec}^2 x\, dy + 2dx = (1 + y\cos 2x)\text{cosec}^2 x\, dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0)+1)^2$ is equal to :

(A) $e^{1/2}$ (B) $e^{-1/2}$ (C) e^{-1} (D) e

Q113. JEE Main 2021 (Online) 20th July Evening Shift

Let y = y(x) satisfies the equation $\frac{dy}{dx} - |A| = 0$, for all x > 0, where $A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}$. If $y(\pi) = \pi + 2$, then the value of $y\left(\frac{\pi}{2}\right)$ is :

(A) $\frac{\pi}{2} + \frac{4}{\pi}$ (B) $\frac{\pi}{2} - \frac{1}{\pi}$

(C) $\frac{3\pi}{2} - \frac{1}{\pi}$ (D) $\frac{\pi}{2} - \frac{4}{\pi}$

Q114. JEE Main 2021 (Online) 20th July Morning Shift

Let y = y(x) be the solution of the differential equation $x\tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx, -1 \le x \le 1$, $y\left(\frac{1}{2}\right) = \frac{\pi}{6}$. Then the area of the region bounded by the curves x = 0, $x = \frac{1}{\sqrt{2}}$ and y = y(x) in the upper half plane is :

(A) $\frac{1}{8}(\pi - 1)$ (B) $\frac{1}{12}(\pi - 3)$ (C) $\frac{1}{4}(\pi - 2)$ (D) $\frac{1}{6}(\pi - 1)$

Q115. JEE Main 2021 (Online) 20th July Morning Shift

Let y = y(x) be the solution of the differential equation $e^x\sqrt{1-y^2}dx + \left(\frac{y}{x}\right)dy = 0$, y(1) = -1. Then the value of $(y(3))^2$ is equal to :

(A) $1 - 4e^3$ (B) $1 - 4e^6$ (C) $1 + 4e^3$ (D) $1 + 4e^6$

Q116. JEE Main 2021 (Online) 18th March Evening Shift

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = (y+1)\left((y+1)e^{x^2/2} - x\right)$, 0 < x < 2.1, with y(2) = 0. Then the value of $\frac{dy}{dx}$ at x = 1 is equal to :

(A) $\frac{e^{5/2}}{(1+e^2)^2}$ (B) $\frac{5e^{1/2}}{(e^2+1)^2}$

(C) $-\frac{2e^2}{(1+e^2)^2}$ (D) $\frac{-e^{3/2}}{(e^2+1)^2}$

Q117. JEE Main 2021 (Online) 17th March Evening Shift

If the curve y = y(x) is the solution of the differential equation

$2(x^2 + x^{5/4})dy - y(x + x^{1/4})dx = 2x^{9/4}dx$, x > 0 which

passes through the point $\left(1, 1 - \frac{4}{3}\log_e 2\right)$, then the value of y(16) is equal to :

(A) $4\left(\frac{31}{3} - \frac{8}{3}\log_e 3\right)$

(B) $\left(\frac{31}{3} - \frac{8}{3}\log_e 3\right)$

(C) $\left(\frac{31}{3} + \frac{8}{3}\log_e 3\right)$

(D) $4\left(\frac{31}{3} + \frac{8}{3}\log_e 3\right)$

Q118. JEE Main 2021 (Online) 17th March Evening Shift

Let y = y(x) be the solution of the differential equation

$\cos x(3\sin x + \cos x + 3)dy = (1 + y\sin x(3\sin x + \cos x + 3))dx, 0 \le x \le \frac{\pi}{2}, y(0) = 0$. Then, $y\left(\frac{\pi}{3}\right)$ is equal to :

(A) $2\log_e\left(\frac{\sqrt{3}+7}{2}\right)$

(B) $2\log_e\left(\frac{3\sqrt{3}-8}{4}\right)$

(C) $2\log_e\left(\frac{2\sqrt{3}+10}{11}\right)$

(D) $2\log_e\left(\frac{2\sqrt{3}+9}{6}\right)$

Q119. JEE Main 2021 (Online) 17th March Morning Shift

Which of the following is true for y(x) that satisfies the differential equation

$\frac{dy}{dx}$ = xy − 1 + x − y; y(0) = 0 :

(A) y(1) = 1

(B) $y(1) = e^{-\frac{1}{2}} - 1$

(C) $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$

(D) $y(1) = e^{\frac{1}{2}} - 1$

Q120. JEE Main 2021 (Online) 16th March Evening Shift

If y = y(x) is the solution of the differential equation

$\frac{dy}{dx}$ + (tan x) y = sin x, $0 \le x \le \frac{\pi}{3}$, with y(0) = 0, then $y\left(\frac{\pi}{4}\right)$ equal to :

(A) $\frac{1}{2}\log_e 2$

(B) $\left(\frac{1}{2\sqrt{2}}\right)\log_e 2$

(C) $\log_e 2$

(D) $\frac{1}{4}\log_e 2$

Q121. JEE Main 2021 (Online) 16th March Evening Shift

Let C_1 be the curve obtained by the solution of differential equation

$2xy\frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the

solution of $\frac{2xy}{x^2-y^2} = \frac{dy}{dx}$. If both the curves pass through (1, 1), then the area enclosed by the curves C_1 and C_2 is equal to :

(A) $\frac{\pi}{4}$ + 1 (B) π + 1 (C) $\pi - 1$ (D) $\frac{\pi}{2} - 1$

Q122. JEE Main 2021 (Online) 16th March Morning Shift

If y = y(x) is the solution of the differential equation,

$\frac{dy}{dx} + 2y\tan x = \sin x, y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of the function y(x) over R is equal to:

(A) $\frac{1}{8}$ (B) 8 (C) $-\frac{15}{4}$ (D) $\frac{1}{2}$

Q123. JEE Main 2021 (Online) 26th February Morning Shift

The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time t = 0. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{k}{\log_e\left(\frac{6}{5}\right)}$ hours, then $\left(\frac{k}{\log_e 2}\right)^2$ is equal to :

(A) 16 (B) 8 (C) 2 (D) 4

Q124. JEE Main 2021 (Online) 25th February Morning Shift

If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point :

(A) (4, 4) (B) (5, 5) (C) (5, 4) (D) (4, 5)

Q125. JEE Main 2021 (Online) 24th February Morning Shift

The population P = P(t) at time 't' of a certain species follows the differential equation

$\frac{dP}{dt}$ = 0.5P – 450. If P(0) = 850, then the time at which population becomes zero is :

(A) $\log_e 18$ (B) $\frac{1}{2}\log_e 18$ (C) $2\log_e 18$ (D) $\log_e 9$

Q126. JEE Main 2020 (Online) 6th September Morning Slot

The general solution of the differential equation

$\sqrt{1+x^2+y^2+x^2y^2}$ + xy$\frac{dy}{dx}$ = 0 is :

(where C is a constant of integration)

(A) $\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right)+C$

(B) $\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right)+C$

(C) $\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right)+C$

(D) $\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right)+C$

Q127. JEE Main 2020 (Online) 5th September Evening Slot

Let y = y(x) be the solution of the differential equation

cosx$\frac{dy}{dx}$ + 2ysinx = sin2x, x $\in \left(0, \frac{\pi}{2}\right)$.

If y$\left(\frac{\pi}{3}\right)$ = 0, then y$\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{1}{\sqrt{2}}-1$ (B) $\sqrt{2}-2$ (C) $2-\sqrt{2}$ (D) $2+\sqrt{2}$

Q128. JEE Main 2020 (Online) 5th September Morning Slot

If y = y(x) is the solution of the differential

equation $\frac{5+e^x}{2+y}.\frac{dy}{dx}+e^x=0$ satisfying y(0) = 1, then a value of y($\log_e$13) is :

(A) -1 (B) 1 (C) 0 (D) 2

Q129. JEE Main 2020 (Online) 4th September Evening Slot

The solution of the differential equation

$\frac{dy}{dx}-\frac{y+3x}{\log_e(y+3x)}+3=0$ is:

(where c is a constant of integration)

(A) $x-\frac{1}{2}(\log_e(y+3x))^2=C$ (B) $y+3x-\frac{1}{2}(\log_e x)^2=C$

(C) x − $\log_e$(y+3x) = C (D) x − 2$\log_e$(y+3x) = C

Q130. JEE Main 2020 (Online) 4th September Morning Slot

Let y = y(x) be the solution of the differential equation,
xy'- y = x^2(xcosx + sinx), x > 0. if y (π) = π then
$y''\left(\frac{\pi}{2}\right)+y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $1+\frac{\pi}{2}$ (B) $2+\frac{\pi}{2}+\frac{\pi^2}{4}$

(C) $2+\frac{\pi}{2}$ (D) $1+\frac{\pi}{2}+\frac{\pi^2}{4}$

Q131. JEE Main 2020 (Online) 3rd September Evening Slot

If x^3dy + xy dx = x^2dy + 2y dx; y(2) = e and
x > 1, then y(4) is equal to :

(A) $\frac{\sqrt{e}}{2}$ (B) $\frac{1}{2}+\sqrt{e}$ (C) $\frac{3}{2}+\sqrt{e}$ (D) $\frac{3}{2}\sqrt{e}$

Q132. JEE Main 2020 (Online) 3rd September Morning Slot

The solution curve of the differential equation,

$(1+e^{-x})(1+y^2)\frac{dy}{dx}=y^2$,

which passes through the point (0, 1), is :

(A) $y^2+1=y\left(\log_e\left(\frac{1+e^{-x}}{2}\right)+2\right)$

(B) $y^2+1=y\left(\log_e\left(\frac{1+e^{x}}{2}\right)+2\right)$

(C) $y^2=1+y\log_e\left(\frac{1+e^{-x}}{2}\right)$

(D) $y^2=1+y\log_e\left(\frac{1+e^{x}}{2}\right)$

Q133. JEE Main 2020 (Online) 2nd September Evening Slot

If a curve y = f(x), passing through the point (1, 2), is the solution of the differential equation,
$2x^2dy=(2xy+y^2)dx$, then $f\left(\frac{1}{2}\right)$ is equal to :

(A) $\frac{1}{1-\log_e 2}$ (B) $\frac{1}{1+\log_e 2}$

(C) $\frac{-1}{1+\log_e 2}$ (D) $1+\log_e 2$

Q134. JEE Main 2020 (Online) 2nd September Morning Slot

Let y = y(x) be the solution of the differential equation,
$\frac{2+\sin x}{y+1}\cdot\frac{dy}{dx}=-\cos x$, y > 0,y(0) = 1.
If y(π) = a and $\frac{dy}{dx}$ at x = π is b, then the ordered pair (a, b) is equal to :

(A) (2, 1) (B) $\left(2,\frac{3}{2}\right)$ (C) (1, -1) (D) (1, 1)

Q135. JEE Main 2020 (Online) 9th January Evening Slot

If $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$; y(1) = 1; then a value of x satisfying y(x) = e is :

(A) $\sqrt{2}e$ (B) $\frac{1}{2}\sqrt{3}e$ (C) $\frac{e}{\sqrt{2}}$ (D) $\sqrt{3}e$

Q136. JEE Main 2020 (Online) 8th January Morning Slot

Let y = y(x) be a solution of the differential equation,

$\sqrt{1-x^2}\frac{dy}{dx} + \sqrt{1-y^2} = 0$, |x| < 1.

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(-\frac{1}{\sqrt{2}}\right)$ is equal to :

(A) $-\frac{\sqrt{3}}{2}$ (B) None of those (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Q137. JEE Main 2020 (Online) 7th January Evening Slot

Let y = y(x) be the solution curve of the differential equation,

$\left(y^2 - x\right)\frac{dy}{dx} = 1$, satisfying y(0) = 1. This curve intersects the x-axis at a point whose abscissa is :

(A) 2 + e (B) -e (C) 2 (D) 2 - e

Q138. JEE Main 2020 (Online) 7th January Morning Slot

If y = y(x) is the solution of the differential equation, $e^y\left(\frac{dy}{dx} - 1\right) = e^x$ such that y(0) = 0, then y(1) is equal to:

(A) $2 + \log_e 2$ (B) $\log_e 2$ (C) $1 + \log_e 2$ (D) 2e

Q139. JEE Main 2019 (Online) 12th April Evening Slot

The general solution of the differential equation $(y^2 - x^3)dx - xydy = 0$ $(x \neq 0)$ is : (where c is a constant of integration)

(A) $y^2 + 2x^3 + cx^2 = 0$ (B) $y^2 + 2x^2 + cx^3 = 0$

(C) $y^2 - 2x + cx^3 = 0$ (D) $y^2 - 2x^3 + cx^2 = 0$

Q140. JEE Main 2019 (Online) 12th April Morning Slot

Consider the differential equation, $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$, If value of y is 1 when x = 1, then the value of x for which y = 2, is :

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $\frac{1}{2} + \frac{1}{\sqrt{e}}$

(C) $\frac{5}{2} + \frac{1}{\sqrt{e}}$ (D) $\frac{3}{2} - \sqrt{e}$

Q141. JEE Main 2019 (Online) 10th April Evening Slot

Let y = y(x) be the solution of the differential equation,
$\frac{dy}{dx} + y\tan x = 2x + x^2\tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that y(0) = 1. Then :

(A) $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$

(B) $y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$

(C) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$

(D) $y'\left(\frac{\pi}{4}\right) + y'\left(-\frac{\pi}{4}\right) = -\sqrt{2}$

Q142. JEE Main 2019 (Online) 10th April Morning Slot

If y = y(x) is the solution of the differential equation
$\frac{dy}{dx} = (\tan x - y)\sec^2 x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,
such that y (0) = 0, then $y\left(-\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{1}{2} - e$ (B) $e - 2$ (C) $2 + \frac{1}{e}$ (D) $\frac{1}{e} - 2$

Q143. JEE Main 2019 (Online) 9th April Evening Slot

If $\cos x\frac{dy}{dx} - y\sin x = 6x$, (0 < x < $\frac{\pi}{2}$)
and $y\left(\frac{\pi}{3}\right)$ = 0 then $y\left(\frac{\pi}{6}\right)$ is equal to :-

(A) $-\frac{\pi^2}{2}$ (B) $-\frac{\pi^2}{4\sqrt{3}}$

(C) $\frac{\pi^2}{2\sqrt{3}}$ (D) $-\frac{\pi^2}{2\sqrt{3}}$

Q144. JEE Main 2019 (Online) 9th April Morning Slot

The solution of the differential equation

$x\frac{dy}{dx} + 2y = x^2$ (x $\neq$ 0) with y(1) = 1, is :

(A) $y = \frac{4}{5}x^3 + \frac{1}{5x^2}$ (B) $y = \frac{3}{4}x^2 + \frac{1}{4x^2}$

(C) $y = \frac{x^2}{4} + \frac{3}{4x^2}$ (D) $y = \frac{x^3}{5} + \frac{1}{5x^2}$

Q145. JEE Main 2019 (Online) 8th April Morning Slot

Let y = y(x) be the solution of the differential equation,

$(x^2+1)^2\frac{dy}{dx} + 2x(x^2+1)y = 1$

such that y(0) = 0. If $\sqrt{a}y(1) = \frac{\pi}{32}$, then the value of 'a' is :

(A) $\frac{1}{2}$ (B) $\frac{1}{16}$ (C) 1 (D) $\frac{1}{4}$

Q146. JEE Main 2019 (Online) 12th January Evening Slot

If a curve passes through the point (1, −2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2-2y}{x}$, then the curve also passes through the point :

(A) (−1, 2) (B) $\left(-\sqrt{2}, 1\right)$ (C) $\left(\sqrt{3}, 0\right)$ (D) (3, 0)

Q147. JEE Main 2019 (Online) 12th January Morning Slot

Let y = y(x) be the solution of the differential equation, $x\frac{dy}{dx}$ + y = x $\log_e$ x, (x > 1). If 2y(2) = $\log_e$ 4 − 1, then y(e) is equal to :

(A) $-\frac{e}{2}$ (B) $-\frac{e^2}{2}$ (C) $\frac{e^2}{4}$ (D) $\frac{e}{4}$

Q148. JEE Main 2019 (Online) 11th January Evening Slot

The solution of the differential equation,

$\frac{dy}{dx}$ = $(x - y)^2$, when y(1) = 1, is :

(A) $-\log_e\left|\frac{1+x-y}{1-x+y}\right|$ = x + y − 2

(B) $\log_e\left|\frac{2-x}{2-y}\right|$ = x − y

(C) $\log_e\left|\frac{2-y}{2-x}\right|$ = 2(y − 1)

(D) $-\log_e\left|\frac{1-x+y}{1+x-y}\right|$ = 2(x − 1)

Q149. JEE Main 2019 (Online) 11th January Morning Slot

If y(x) is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x},\ x > 0,$ where $y(1) = \frac{1}{2}e^{-2}$, then

(A) y($\log_e$2) = $\log_e$4

(B) y(x) is decreasing in (0, 1)

(C) y($\log_e$2) = $\frac{\log_e 2}{4}$

(D) y(x) is decreasing in $\left(\frac{1}{2}, 1\right)$

Q150. JEE Main 2019 (Online) 10th January Evening Slot

The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2)dx + 2xy\,dy = 0$ which passes through (1, 1) is :

(A) a circle with centre on the y-axis

(B) an ellipse with major axis along the y-axis

(C) a circle with centre on the x-axis

(D) a hyperbola with transverse axis along the x-axis

Q151. JEE Main 2019 (Online) 10th January Evening Slot

Let f be a differentiable function such that f '(x) = 7 - $\frac{3}{4}\frac{f(x)}{x}$, (x > 0) and f(1) $\neq$ 4. Then $\lim_{x\to 0^+} xf\left(\frac{1}{x}\right)$:

(A) does not exist

(B) exists and equals $\frac{4}{7}$

(C) exists and equals 4

(D) exists and equals 0

Q152. JEE Main 2019 (Online) 10th January Morning Slot

If $\frac{dy}{dx} + \frac{3}{\cos^2 x}y = \frac{1}{\cos^2 x}$, $x \in \left(\frac{-\pi}{3}, \frac{\pi}{3}\right)$ and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals -

(A) $\frac{1}{3} + e^6$

(B) $\frac{1}{3}$

(C) $\frac{1}{3}$ + e^3

(D) $-\frac{4}{3}$

Q153. JEE Main 2019 (Online) 9th January Evening Slot

Let f : [0,1] $\rightarrow$ **R** be such that f(xy) = f(x).f(y), for all x, y $\in$ [0, 1], and f(0) $\neq$ 0. If y = y(x) satiesfies the differential equation, $\frac{dy}{dx}$ = f(x) with y(0) = 1, then y$\left(\frac{1}{4}\right)$ + y$\left(\frac{3}{4}\right)$ is equal to :

(A) 3

(B) 4

(C) 2

(D) 5

Q154. JEE Main 2019 (Online) 9th January Morning Slot

If y = y(x) is the solution of the differential equation,

x$\frac{dy}{dx}$ + 2y = x^2, satisfying y(1) = 1, then y($\frac{1}{2}$) is equal to :

(A) $\frac{7}{64}$

(B) $\frac{49}{16}$

(C) $\frac{1}{4}$

(D) $\frac{13}{16}$

Q155. JEE Main 2018 (Offline)

Let y = y(x) be the solution of the differential equation

$\sin x\frac{dy}{dx} + y\cos x = 4x, x \in (0, \pi)$.

If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to :

(A) $-\frac{4}{9}\pi^2$

(B) $\frac{4}{9\sqrt{3}}\pi^2$

(C) $-\frac{8}{9\sqrt{3}}\pi^2$

(D) $-\frac{8}{9}\pi^2$

Q156. JEE Main 2018 (Online) 15th April Evening Slot

The curve satisfying the differential equation, ($x^2 - y^2$) dx + 2xydy = 0 and passing through the point (1, 1) is :

(A) a circle of radius one.

(B) a hyperbola.

(C) an ellipse.

(D) a circle of radius two.

Q157. JEE Main 2018 (Online) 15th April Morning Slot

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$,

where $f(x) = \begin{cases} 1, & x \in [0,1] \\ 0, & \text{otherwise} \end{cases}$

If y(0) = 0, then $y\left(\frac{3}{2}\right)$ is :

(A) $\frac{e^2+1}{2e^4}$ (B) $\frac{1}{2e}$ (C) $\frac{e^2-1}{e^3}$ (D) $\frac{e^2-1}{2e^3}$

Q158. JEE Main 2017 (Online) 9th April Morning Slot

If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ and

$(x^2 - 1)\frac{d^2y}{dx^2} + \lambda x \frac{dy}{dx} + ky = 0$,

then $\lambda + k$ is equal to :

(A) -23 (B) -24 (C) 26 (D) -26

Q159. JEE Main 2017 (Online) 8th April Morning Slot

The curve satisfying the differential equation, $ydx - (x + 3y^2)dy = 0$ and passing through the point (1, 1), also passes through the point :

(A) $\left(\frac{1}{4}, -\frac{1}{2}\right)$ (B) $\left(-\frac{1}{3}, \frac{1}{3}\right)$

(C) $\left(\frac{1}{3}, -\frac{1}{3}\right)$ (D) $\left(\frac{1}{4}, \frac{1}{2}\right)$

Q160. JEE Main 2017 (Offline)

If $(2 + \sin x)\frac{dy}{dx} + (y + 1)\cos x = 0$ and y(0) = 1,

then $y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $-\frac{2}{3}$ (B) $-\frac{1}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

Q161. JEE Main 2016 (Online) 10th April Morning Slot

The solution of the differential equation

$\frac{dy}{dx} + \frac{y}{2}\sec x = \frac{\tan x}{2y}$,

where $0 \le x < \frac{\pi}{2}$, and y (0) = 1, is given by :

(A) $y = 1 - \frac{x}{\sec x + \tan x}$ (B) $y^2 = 1 + \frac{x}{\sec x + \tan x}$

(C) $y^2 = 1 - \frac{x}{\sec x + \tan x}$ (D) $y = 1 + \frac{x}{\sec x + \tan x}$

Q162. JEE Main 2016 (Online) 9th April Morning Slot

If f(x) is a differentiable function in the interval $(0, \infty)$ such that f (1) = 1 and

$\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$, for each x > 0, then $f\left(\frac{3}{2}\right)$ equal to :

(A) $\frac{13}{6}$ (B) $\frac{23}{18}$ (C) $\frac{25}{9}$ (D) $\frac{31}{18}$

Q163. JEE Main 2016 (Offline)

If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy)dx = x\, dy$, then $f\left(-\frac{1}{2}\right)$ is equal to :

(A) $\frac{2}{5}$ (B) $\frac{4}{5}$ (C) $-\frac{2}{5}$ (D) $-\frac{4}{5}$

Q164. JEE Main 2015 (Offline)

Let $y(x)$ be the solution of the differential equation

$(x \log x)\frac{dy}{dx} + y = 2x \log x, (x \geq 1)$. Then $y(e)$ is equal to :

(A) 2 (B) $2e$ (C) e (D) 0

Q165. JEE Main 2014 (Offline)

Let the population of rabbits surviving at time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals:

(A) $600 - 500\, e^{t/2}$ (B) $400 - 300\, e^{-t/2}$ (C) $400 - 300\, e^{t/2}$ (D) $300 - 200\, e^{-t/2}$

Q166. JEE Main 2013 (Offline)

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is

(A) 2500 (B) 3000 (C) 3500 (D) 4500

Q167. AIEEE 2012

The population $p(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5\, p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

(A) $2ln\ 18$ (B) $ln\ 9$ (C) $\frac{1}{2}ln\ 18$ (D) $ln\ 18$

Q168. AIEEE 2011

Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is

(A) $I - \frac{kT^2}{2}$ (B) $I - \frac{k(T-t)^2}{2}$

(C) e^{-kT} (D) $T^2 - \frac{1}{k}$

Q169. AIEEE 2011

If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :

(A) 5 (B) 13 (C) -2 (D) 7

Q170. AIEEE 2010

Solution of the differential equation

$\cos x\, dy = y(\sin x - y)dx,\ 0 < x < \frac{\pi}{2}$ is :

(A) $y \sec x = \tan x + c$ (B) $y \tan x = \sec x + c$

(C) $\tan x = (\sec x + c)y$ (D) $\sec x = (\tan x + c)y$

Q171. AIEEE 2008

The solution of the differential equation

$\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is :

(A) $y = \ln x + x$ (B) $y = x \ln x + x^2$

(C) $y = xe^{(x-1)}$ (D) $y = x\ \ln x + x$

Q172. AIEEE 2006

The differential equation whose solution is $Ax^2 + By^2 = 1$
where A and B are arbitrary constants is of

(A) second order and second degree (B) first order and second degree

(C) first order and first degree (D) second order and first degree

Q173. AIEEE 2005

The differential equation representing the family of curves $y^2 = 2c\left(x + \sqrt{c}\right)$, where $c > 0$, is a parameter, is of order and degree as follows:

(A) order 1, degree 2 (B) order 1, degree 1 (C) order 1, degree 3 (D) order 2, degree 2

Q174. AIEEE 2005

If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is :

(A) $y\log\left(\frac{x}{y}\right) = cx$ (B) $x\log\left(\frac{y}{x}\right) = cy$

(C) $\log\left(\frac{y}{x}\right) = cx$ (D) $\log\left(\frac{x}{y}\right) = cy$

Q175. AIEEE 2004

Solution of the differential equation $ydx + (x + x^2y)dy = 0$ is

(A) $log\, y = Cx$ (B) $-\frac{1}{xy} + \log y = C$ (C) $\frac{1}{xy} + \log y = C$ (D) $-\frac{1}{xy} = C$

Q176. AIEEE 2003

The solution of the differential equation

$(1 + y^2) + \left(x - e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0$, is :

(A) $xe^{2\tan^{-1}y} = e^{\tan^{-1}y} + k$ (B) $(x - 2) = ke^{2\tan^{-1}y}$

(C) $2xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + k$ (D) $xe^{\tan^{-1}y} = \tan^{-1}y + k$

Q177. AIEEE 2003

The degree and order of the differential equation of the family of all parabolas whose axis is x-axis, are respectively.

(A) $2, 3$ (B) $2, 1$ (C) $1, 2$ (D) $3, 2.$

Q178. AIEEE 2002

The order and degree of the differential equation
$\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are

(A) $\left(1, \frac{2}{3}\right)$ (B) $(3, 1)$ (C) $(3, 3)$ (D) $(1, 2)$

Q179. AIEEE 2002

The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$

(A) $\frac{e^{-2x}}{4}$ (B) $\frac{e^{-2x}}{4} + cx + d$

(C) $\frac{1}{4}e^{-2x} + cx^2 + d$ (D) $\frac{1}{4}e^{-4x} + cx + d$

Numerical 50 questions

Q180. JEE Main 2026 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x - \sqrt{x^2-1}\right) - x\right)dx = 0, x \geq 1$. If $y(1) = 1$, then the greatest integer less than $y(\sqrt{5})$ is ____ .

Q181. JEE Main 2026 (Online) 5th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(\tan x)^{1/2}\, dy = \left(\sec^3 x - (\tan x)^{3/2} y\right)dx, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then α^4 equals ____ .

Q182. JEE Main 2026 (Online) 5th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $x \sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx, y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y(e^{12})}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \leq 6$, is ____ .

Q183. JEE Main 2026 (Online) 4th April Evening Shift

Let f be a twice differentiable function such that

$f(x) = \int_0^x \tan(t-x)dt - \int_0^x f(t)\tan t dt, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to _____.

Q184. JEE Main 2026 (Online) 23rd January Evening Shift

If the solution curve $y = f(x)$ of the differential equation

$\left(x^2 - 4\right)y' - 2xy + 2x\left(4 - x^2\right)^2 = 0, x > 2,$

passes through the point $(3, 15)$, then the local maximum value of f is ____

Q185. JEE Main 2026 (Online) 23rd January Morning Shift

Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right)\mathrm{dt}$. Then the mean of $f(\log_e(1)), f(\log_e(2)), \ldots\ldots, f(\log_e(625))$ is equal to ____ .

Q186. JEE Main 2025 (Online) 2nd April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x \cdot \sec^2 x$ such that $y(0) = \frac{5}{4}$. Then $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ is equal to_____

Q187. JEE Main 2025 (Online) 28th January Evening Shift

If $y = y(x)$ is the solution of the differential equation, $\sqrt{4-x^2}\frac{dy}{dx} = \left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2 - y\right)\sin^{-1}\left(\frac{x}{2}\right), -2 \le x \le 2, y(2) = \frac{\pi^2-8}{4}$, then $y^2(0)$ is equal to _____.

Q188. JEE Main 2025 (Online) 24th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$2\cos x\frac{dy}{dx} = \sin 2x - 4y\sin x, x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right) = 0$, then $y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ is equal to _____.

Q189. JEE Main 2025 (Online) 24th January Morning Shift

Let f be a differentiable function such that $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt, x \ge 0$. Then $f(2)$ is equal to _____ .

Q190. JEE Main 2025 (Online) 22nd January Evening Shift

Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$ such that $f(0) = 0$. If $6\int_{-1/2}^{1/2} f(x)dx = 2\pi - \alpha$ then α^2 is equal to _____ .

Q191. JEE Main 2024 (Online) 9th April Evening Shift

For a differentiable function $f : \mathbb{R} \to \mathbb{R}$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in \mathbb{R}$, $f(0) = 1$ and $\lim\limits_{x\to-\infty} f(x) = 7$. Then $9f(-\log_e 3)$ is equal to _____.

Q192. JEE Main 2024 (Online) 8th April Evening Shift

Let $\alpha|x| = |y|e^{xy-\beta}, \alpha, \beta \in \mathbf{N}$ be the solution of the differential equation $x\,dy - y\,dx + xy(x\,dy + y\,dx) = 0, y(1) = 2$. Then $\alpha + \beta$ is equal to _____

Q193. JEE Main 2024 (Online) 6th April Evening Shift

If the solution $y(x)$ of the given differential equation $(e^y + 1)\cos x\,dx + e^y \sin x\,dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____.

Q194. JEE Main 2024 (Online) 5th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} + \frac{2x}{(1+x^2)^2}y = xe^{\frac{1}{(1+x^2)}}; y(0) = 0.$

Then the area enclosed by the curve $f(x) = y(x)e^{-\frac{1}{(1+x^2)}}$ and the line $y - x = 4$ is _____.

Q195. JEE Main 2024 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2 dx = dy, y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}, \gamma, \delta \in \mathbb{Z}$, then $\gamma + \delta$ equals _____.

Q196. JEE Main 2024 (Online) 4th April Morning Shift

Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4\sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____.

Q197. JEE Main 2024 (Online) 1st February Evening Shift

If $\frac{dx}{dy} = \frac{1+x-y^2}{y}, x(1) = 1$, then $5x(2)$ is equal to _____.

Q198. JEE Main 2024 (Online) 1st February Morning Shift

If $x = x(t)$ is the solution of the differential equation $(t+1)dx = \left(2x + (t+1)^4\right)dt, x(0) = 2$, then, $x(1)$ equals _____.

Q199. JEE Main 2024 (Online) 31st January Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\sec^2 x dx + \left(e^{2y}\tan^2 x + \tan x\right)dy = 0, 0 < x < \frac{\pi}{2}, y(\pi/4) = 0.$

If $y(\pi/6) = \alpha$, then $e^{8\alpha}$ is equal to _____.

Q200. JEE Main 2024 (Online) 30th January Evening Shift

Let $Y = Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)} + 1, Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____.

Q201. JEE Main 2024 (Online) 30th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(1-x^2\right)\mathrm{d}y = \left[xy + \left(x^3+2\right)\sqrt{3\left(1-x^2\right)}\right]\mathrm{d}x, -1 < x < 1, y(0) = 0$. If $y\left(\frac{1}{2}\right) = \frac{\mathrm{m}}{\mathrm{n}}$, m and n are co-prime numbers, then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Q202. JEE Main 2024 (Online) 29th January Morning Shift

If the solution curve $y = y(x)$ of the differential equation $\left(1+y^2\right)\left(1+\log_{\mathrm{e}} x\right)dx + xdy = 0, x > 0$ passes through the point $(1,1)$ and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is _____.

Q203. JEE Main 2024 (Online) 27th January Evening Shift

If the solution curve, of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+y-2}{x-y}$ passing through the point $(2,1)$ is $\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{\beta}\log_{\mathrm{e}}\left(\alpha + \left(\frac{y-1}{x-1}\right)^2\right) = \log_{\mathrm{e}}|x-1|$, then $5\beta + \alpha$ is equal to _____.

Q204. JEE Main 2024 (Online) 27th January Morning Shift

If the solution of the differential equation

$(2x+3y-2)\mathrm{d}x + (4x+6y-7)\mathrm{d}y = 0, y(0) = 3$, is

$\alpha x + \beta y + 3\log_e|2x+3y-\gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____.

Q205. JEE Main 2023 (Online) 13th April Evening Shift

If $y = y(x)$ is the solution of the differential equation

$\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}, x > 1$ such that

$y(2) = \frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2}) = \alpha\log_e(\sqrt{\alpha}+\beta) + \beta - \sqrt{\gamma}, \alpha, \beta, \gamma \in \mathbb{N}$, then $\alpha\beta\gamma$ is equal to :

Q206. JEE Main 2023 (Online) 10th April Evening Shift

Let the tangent at any point P on a curve passing through the points (1, 1) and $\left(\frac{1}{10}, 100\right)$, intersect positive x-axis and y-axis at the points A and B respectively. If $\mathrm{PA} : \mathrm{PB} = 1 : k$ and $y = y(x)$ is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$, then $4y(1) - 6\log_{\mathrm{e}} 3$ is equal to _____.

Q207. JEE Main 2023 (Online) 8th April Evening Shift

Let the solution curve $x = x(y), 0 < y < \frac{\pi}{2}$, of the differential equation $\left(\log_e(\cos y)\right)^2\cos y\,\mathrm{d}x - (1 + 3x\log_e(\cos y))\sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$. If $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where m and n are coprime, then mn is equal to _____.

Q208. JEE Main 2023 (Online) 8th April Morning Shift

If the solution curve of the differential equation $(y-2\log_e x)dx+\left(x\log_e x^2\right)dy=0, x>1$ passes through the points $\left(e,\frac{4}{3}\right)$ and $\left(e^4,\alpha\right)$, then α is equal to _____.

Q209. JEE Main 2023 (Online) 6th April Morning Shift

Let $y=y(x)$ be a solution of the differential equation $(x\cos x)dy+(xy\sin x+y\cos x-1)dx=0, 0<x<\frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right)=\sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right)+2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____.

Q210. JEE Main 2022 (Online) 27th July Morning Shift

Let $y=y(x)$ be the solution curve of the differential equation $\sin\left(2x^2\right)\log_e\left(\tan x^2\right)dy+\left(4xy-4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)\right)dx=0, 0<x<\sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}},1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to _____.

Q211. JEE Main 2022 (Online) 26th July Evening Shift

Suppose $y=y(x)$ be the solution curve to the differential equation $\frac{dy}{dx}-y=2-e^{-x}$ such that $\lim\limits_{x\to\infty}y(x)$ is finite. If a and b are respectively the x - and y-intercepts of the tangent to the curve at $x=0$, then the value of $a-4b$ is equal to _____.

Q212. JEE Main 2022 (Online) 25th July Evening Shift

Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}=\frac{4y^3+2yx^2}{3xy^2+x^3}, y(1)=1$.

If for some $n\in\mathbb{N}, y(2)\in[n-1,n)$, then n is equal to _____.

Q213. JEE Main 2022 (Online) 30th June Morning Shift

A hostel has 100 students. On a certain day (consider it day zero) it was found that two students are infected with some virus. Assume that the rate at which the virus spreads is directly proportional to the product of the number of infected students and the number of non-infected students. If the number of infected students on 4th day is 30, then number of infected students on 8th day will be _____.

Q214. JEE Main 2022 (Online) 29th June Evening Shift

Let y = y(x), x > 1, be the solution of the differential equation $(x-1)\frac{dy}{dx}+2xy=\frac{1}{x-1}$, with $y(2)=\frac{1+e^4}{2e^4}$. If $y(3)=\frac{e^\alpha+1}{\beta e^\alpha}$, then the value of $\alpha+\beta$ is equal to _____.

Q215. JEE Main 2022 (Online) 29th June Morning Shift

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}$, $0 < x < \frac{\pi}{2}$ with $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}$. If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to _____.

Q216. JEE Main 2022 (Online) 27th June Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(1-x^2)dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx, -1 < x < 1$, and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}y(x)dx = k$, then k $^{-1}$ is equal to _____.

Q217. JEE Main 2022 (Online) 26th June Morning Shift

Let the solution curve y = y(x) of the differential equation

$(4+x^2)dy - 2x(x^2+3y+4)dx = 0$ pass through the origin. Then y(2) is equal to _____.

Q218. JEE Main 2022 (Online) 26th June Morning Shift

Let $S = (0, 2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$. Let $y = y(x)$, x ∈ S, be the solution curve of the differential equation $\frac{dy}{dx} = \frac{1}{1+\sin 2x}$, $y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve y = y(x) with the curve $y = \sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to _____.

Q219. JEE Main 2021 (Online) 27th July Evening Shift

Let y = y(x) be the solution of the differential equation dy = $e^{\alpha x + y}$ dx; $\alpha \in$ N. If y($\log_e 2$) = $\log_e 2$ and y(0) = $\log_e\left(\frac{1}{2}\right)$, then the value of α is equal to _____.

Q220. JEE Main 2021 (Online) 27th July Morning Shift

If $y = y(x), y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation $\sec y\frac{dy}{dx} - \sin(x+y) - \sin(x-y) = 0$, with y(0) = 0, then $5y'\left(\frac{\pi}{2}\right)$ is equal to _____.

Q221. JEE Main 2021 (Online) 25th July Morning Shift

Let y = y(x) be solution of the following differential equation $e^y\frac{dy}{dx} - 2e^y\sin x + \sin x\cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$ If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha+\beta)$ is equal to _____.

Q222. JEE Main 2021 (Online) 22th July Evening Shift

Let y = y(x) be the solution of the differential equation $\left((x+2)e^{\left(\frac{y+1}{x+2}\right)} + (y+1)\right)dx = (x+2)dy$, y(1) = 1. If the domain of y = y(x) is an open interval (α, β), then | α + β| is equal to _____.

Q223. JEE Main 2021 (Online) 20th July Evening Shift

Let a curve y = y(x) be given by the solution of the differential equation $\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x}-1}dy$. If it intersects y-axis at y = -1, and the intersection point of the curve with x-axis is (α, 0), then e^{α} is equal to _____.

Q224. JEE Main 2021 (Online) 18th March Evening Shift

Let y = y(x) be the solution of the differential equation

$xdy - ydx = \sqrt{(x^2-y^2)}dx$, $x \geq 1$, with y(1) = 0. If the area bounded by the line x = 1, x = e^{π}, y = 0 and y = y(x) is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to _____.

Q225. JEE Main 2021 (Online) 16th March Morning Shift

Let the curve y = y(x) be the solution of the differential equation, $\frac{dy}{dx}$ = 2(x + 1). If the numerical value of area bounded by the curve y = y(x) and x-axis is $\frac{4\sqrt{8}}{3}$, then the value of y(1) is equal to _____.

Q226. JEE Main 2021 (Online) 26th February Morning Shift

The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$, a > 0 is _____.

Q227. JEE Main 2021 (Online) 26th February Morning Shift

If y = y(x) is the solution of the equation

$e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x$, y(0) = 0; then

$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to _____.

Q228. JEE Main 2021 (Online) 25th February Evening Shift

If the curve, y = y(x) represented by the solution of the differential equation $(2xy^2 - y)dx + xdy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then |y(1)| is equal to _____.

Q229. JEE Main 2020 (Online) 9th January Morning Slot

If for $x \geq 0$, y = y(x) is the solution of the differential equation $(x + 1)dy = ((x + 1)^2 + y - 3)dx$, y(2) = 0, then y(3) is equal to _____.

Out of syllabus 21 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q230. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

The slope of tangent at any point (x, y) on a curve $y = y(x)$ is $\frac{x^2+y^2}{2xy}, x > 0$. If $y(2) = 0$, then a value of $y(8)$ is :

(A) $-4\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{3}$ (D) $-2\sqrt{3}$

Q231. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

Let a differentiable function f satisfy $f(x) + \int_3^x \frac{f(t)}{t}dt = \sqrt{x+1}, x \geq 3$. Then $12f(8)$ is equal to :

(A) 19 (B) 34 (C) 17 (D) 1

Q232. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

The differential equation of the family of circles passing through the points $(0, 2)$ and $(0, -2)$ is :

(A) $2xy\frac{dy}{dx} + \left(x^2 - y^2 + 4\right) = 0$ (B) $2xy\frac{dy}{dx} + \left(x^2 + y^2 - 4\right) = 0$

(C) $2xy\frac{dy}{dx} + \left(y^2 - x^2 + 4\right) = 0$ (D) $2xy\frac{dy}{dx} - \left(x^2 - y^2 + 4\right) = 0$

Q233. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to

(A) $2\log_e 2$ (B) 4 (C) 1 (D) $4\log_e 2$

Q234. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, $a, b, c \in R$, represents a circle with center (α, β). Then, $\alpha + 2\beta$ is equal to :

(A) -1 (B) 0 (C) 1 (D) 2

Q235. JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, where a, b, c are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11, 6) from this circle is :

(A) 10 (B) 8 (C) 7 (D) 5

Q236. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

The slope of normal at any point (x, y), x > 0, y > 0 on the curve y = y(x) is given by $\frac{x^2}{xy-x^2y^2-1}$. If the curve passes through the point (1, 1), then e . y(e) is equal to

(A) $\frac{1-\tan(1)}{1+\tan(1)}$ (B) tan(1)

(C) 1 (D) $\frac{1+\tan(1)}{1-\tan(1)}$

Q237. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point (2, −3) from the line 3x + 4y = 5, is given by :

(A) $10\frac{d^2y}{dx^2} = 11$ (B) $11\frac{d^2x}{dy^2} = 10$ (C) $10\frac{d^2x}{dy^2} = 11$ (D) $11\frac{d^2y}{dx^2} = 10$

Q238. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

The differential equation satisfied by the system of parabolas

$y^2 = 4a(x + a)$ is :

(A) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

(B) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

(C) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(D) $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

Q239. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

If a curve y = f(x) passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x)dx = \frac{62}{5}$?

(A) $\frac{31}{5}$ (B) 10 (C) 5 (D) $\frac{62}{5}$

Q240. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

If $y = \left(\frac{2}{\pi}x - 1\right)cosec\, x$ is the solution of the differential equation,

$\frac{dy}{dx} + p(x)y = \frac{2}{\pi}cosec\, x$,

$0 < x < \frac{\pi}{2}$, then the function p(x) is equal to :

(A) cot x (B) sec x (C) tan x (D) cosec x

Q241. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b \in R$, is :

(A) $x(y')^2 = x - 2yy'$ (B) $x(y')^2 = 2yy' - x$

(C) $xy'' = y'$ (D) $x(y')^2 = x + 2yy'$

Q242. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

Given that the slope of the tangent to a curve y = y(x) at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the centre of the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is :

(A) $x \log_e|y| = 2(x - 1)$

(B) $x^2 \log_e|y| = -2(x - 1)$

(C) $x \log_e|y| = x - 1$

(D) $x \log_e|y| = -2(x - 1)$

Q243. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The differential equation representing the family of ellipse having foci either on the x-axis or on the y-axis, center at the origin and passing through the point (0, 3) is :

(A) $xy\,y'' + x(y')^2 - y\,y' = 0$

(B) $x + y\,y'' = 0$

(C) $xy\,y' + y^2 - 9 = 0$

(D) $xy\,y' - y^2 + 9 = 0$

Q244. AIEEE 2009 · MCQ (Single Correct Answer)

The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 , and c_2 are arbitrary constants, is

(A) $y'' = y'y$

(B) $yy'' = y'$

(C) $yy'' = (y')^2$

(D) $y' = y^2$

Q245. AIEEE 2008 · MCQ (Single Correct Answer)

The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is :

(A) $(x-2)y'^2 = 25 - (y-2)^2$

(B) $(y-2)y'^2 = 25 - (y-2)^2$

(C) $(y-2)^2y'^2 = 25 - (y-2)^2$

(D) $(x-2)^2y'^2 = 25 - (y-2)^2$

Q246. AIEEE 2007 · MCQ (Single Correct Answer)

The differential equation of all circles passing through the origin and having their centres on the x-axis is :

(A) $y^2 = x^2 + 2xy\frac{dy}{dx}$

(B) $y^2 = x^2 - 2xy\frac{dy}{dx}$

(C) $x^2 = y^2 + xy\frac{dy}{dx}$

(D) $x^2 = y^2 + 3xy\frac{dy}{dx}$

Q247. AIEEE 2004 · MCQ (Single Correct Answer)

The differential equation for the family of circle $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is :

(A) $\left(x^2 + y^2\right)y' = 2xy$

(B) $2\left(x^2 + y^2\right)y' = xy$

(C) $\left(x^2 - y^2\right)y' = 2xy$

(D) $2\left(x^2 - y^2\right)y' = xy$

Q248. JEE Main 2022 (Online) 26th July Morning Shift · **Numerical**

Let a curve $y = y(x)$ pass through the point $(3, 3)$ and the area of the region under this curve, above the x-axis and between the abscissae 3 and $x(> 3)$ be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to _____.

Q249. JEE Main 2021 (Online) 27th August Morning Shift · **Numerical**

If $y^{1/4} + y^{-1/4} = 2x$, and

$(x^2 - 1)\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + \beta y = 0$, then | $\alpha - \beta$ | is equal to _____.

Q250. JEE Main 2021 (Online) 25th July Evening Shift · **Numerical**

Let a curve y = f(x) pass through the point (2, $(\log_e 2)^2$) and have slope $\frac{2y}{x\log_e x}$ for all positive real value of x. Then the value of f(e) is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **C**	3. **A**	4. **D**	5. **C**
6. **B**	7. **C**	8. **D**	9. **C**	10. **D**
11. **D**	12. *Scan QR*	13. **B**	14. **A**	15. **B**
16. **B**	17. **B**	18. **C**	19. **C**	20. **A**
21. **A**	22. **B**	23. **D**	24. **A**	25. **A**
26. **D**	27. **B**	28. **C**	29. **D**	30. **C**
31. **A**	32. **D**	33. **A**	34. **C**	35. **D**
36. **B**	37. **D**	38. **C**	39. **A**	40. **B**
41. **D**	42. **C**	43. **D**	44. **D**	45. **B**
46. **D**	47. **A**	48. **C**	49. **C**	50. **B**
51. **D**	52. **B**	53. **A**	54. **B**	55. **B**
56. **A**	57. **C**	58. **A**	59. **A**	60. **A**
61. **D**	62. **A**	63. **A**	64. **D**	65. **D**
66. **C**	67. **B**	68. **D**	69. **B**	70. **A**
71. **A**	72. **D**	73. **C**	74. **A**	75. **B**
76. **B**	77. **D**	78. **B**	79. **A**	80. **A**
81. **B**	82. **A**	83. **B**	84. **A**	85. **C**
86. **A**	87. **D**	88. **B**	89. **A**	90. **A**
91. **B**	92. **D**	93. **D**	94. **C**	95. **B**
96. **B**	97. **C**	98. **A**	99. **D**	100. **A**
101. **B**	102. **B**	103. **D**	104. **A**	105. **C**
106. **C**	107. **D**	108. **B**	109. **A**	110. **A**
111. **D**	112. **C**	113. **A**	114. **A**	115. **B**
116. **D**	117. **A**	118. **C**	119. **B**	120. **B**
121. **D**	122. **A**	123. **D**	124. **B**	125. **C**
126. **C**	127. **B**	128. **A**	129. **A**	130. **C**
131. **D**	132. **D**	133. **B**	134. **D**	135. **D**
136. **C**	137. **D**	138. **C**	139. **A**	140. **A**
141. **B**	142. **B**	143. **D**	144. **C**	145. **B**
146. **C**	147. **D**	148. **D**	149. **D**	150. **C**

151. **C**	152. **A**	153. **A**	154. **B**	155. **D**
156. **A**	157. **D**	158. **B**	159. **B**	160. **D**
161. **C**	162. **D**	163. **B**	164. **A**	165. **C**
166. **C**	167. **A**	168. **A**	169. **D**	170. **D**
171. **D**	172. **D**	173. **C**	174. **C**	175. **B**
176. **C**	177. **C**	178. **C**	179. **B**	

Numerical

180. **3**	181. **48**	182. **11**	183. **5**	184. **16**
185. **1565**	186. **21**	187. **4**	188. **1**	189. **19**
190. **27**	191. **61**	192. **4**	193. **3**	194. **18**
195. **31**	196. **7**	197. **5**	198. **14**	199. **9**
200. **20**	201. **97**	202. **3**	203. **11**	204. **29**
205. **6**	206. **4**	207. **12**	208. **3**	209. **2**
210. **1**	211. **3**	212. **3**	213. **90**	214. **14**
215. **2**	216. **320**	217. **12**	218. **42**	219. **2**
220. **2**	221. **4**	222. **4**	223. **2**	224. **4**
225. **2**	226. **2**	227. **1**	228. **1**	229. **3**

Out of syllabus

230. **C**	231. **C**	232. **A**	233. **B**	234. **C**
235. **B**	236. **D**	237. **D**	238. **C**	239. **B**
240. **A**	241. **D**	242. **A**	243. **D**	244. **C**
245. **C**	246. **A**	247. **C**	248. **6**	249. **17**
250. **1**				